JavaScript DOM Assignment

Course: Web & App Development – Full Stack Course

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Part 1 – Conceptual Questions

Q1. What is DOM?

Explain **DOM (Document Object Model)** in your own words. Also explain how JavaScript uses DOM to interact with HTML elements.

Q2. What is DOM Tree Structure?

Explain how HTML elements are arranged in a **tree structure**.

Example:

```
HTML
 ├── Head
 |    └── Title
 └── Body
      ├── H1
      └── P
```

Explain the relationship between:

- Parent
- Child
- Sibling elements

Q3. Why do we learn DOM?

Explain why **DOM is important in JavaScript**.

Your answer must include:

- Advantages of using DOM
- Disadvantages of not using DOM
- Difference between **with DOM and without DOM**
- One **real-world example** where DOM is used on websites

Q4. What is the difference between querySelector() and querySelectorAll()?

Explain the difference between these two DOM selection methods.

Your answer must include:

- Explanation
- Coding examples

- One **real-life example**

Q5. Write the difference between the following DOM methods.

Your answer must include:

- Definition
- Syntax
- One **real-life example**

Method

getElementById()

getElementsByClassName()

getElementsByTagName()

Q6. Write the difference between the following DOM properties.

Your answer must include:

- Definition
- Syntax
- One **real-life example**

Property

innerHTML

innerText

textContent

Append

Append Child

Part 2 – MCQs

Q1

What does **DOM** stand for?

A) Document Object Model
B) Data Object Method
C) Document Order Model
D) Data Output Method

Q2

Which object represents the entire HTML document?

A) window
B) document

C) html
D) body

Q3

Which method selects an element using **ID**?

A) getElementsByClassName()
B) querySelector()
C) getElementById()
D) createElement()

Q4

Which method selects elements using **class name**?

A) getElementsByClassName()
B) getElementById()
C) querySelectorAll()
D) appendChild()

Q5

Which method returns **only the first matching element**?

A) querySelector()
B) querySelectorAll()

C) getElementsByTagName()
D) getElementsByClassName()

Q6

Which property allows adding **HTML tags inside content**?

A) innerText
B) textContent
C) innerHTML
D) value

Q7

Which method is used to **create a new HTML element**?

A) appendChild()
B) createElement()
C) removeChild()
D) querySelector()

Q8

Which method is used to **add an element as the last child**?

A) prepend()
B) appendChild()

C) remove()
D) replaceChild()

Q9

Which method removes a **specific child element**?

A) remove()
B) removeChild()
C) deleteChild()
D) pop()

Q10

Which method replaces an existing element with another element?

A) append()
B) createElement()
C) replaceChild()
D) querySelector()

Part 3 – Practical Tasks

Task 1 – Change Text

Create a webpage that contains a paragraph and a button.
When the user clicks the button, the text of the paragraph should change.

Task 2 – Add List Item

Create a webpage that contains a list and a button.
When the user clicks the button, a new list item should be added to the list.

Task 3 – Remove List Item

Create a webpage that contains a list and a button.
When the user clicks the button, the last item of the list should be removed.

Task 4 – Replace Element

Create a webpage that contains a heading and a button.
When the user clicks the button, the old heading should be replaced with a new heading.